Radio essays

collected by Rahma Ammar

Three main factors determine the technical quality of chest x ray

- Inspiration
- Centering
- Penetration

Disease of the chest with normal chest radiograph

- **Obstructive airway** disease(asthma and acute bronchiolitis)
- Early **bronchiectasis**
- **Small** pulmonary lesion and **endobronchial** lesion
- Pulmonary **emboli without infarction**
- Some types of infection(early PCP and early miliary TB)
- **Dry pleurisy** and small pleural effusion
- Mediastinal masses

Solitary pulmonary nodule (size)

A solitary mass **over 4 cm** (**without calcification**) is nearly always either

- primary carcinoma.
- lung abscess .
- **hydatid** .
- or rarely spherical pneumonia.

(benign)

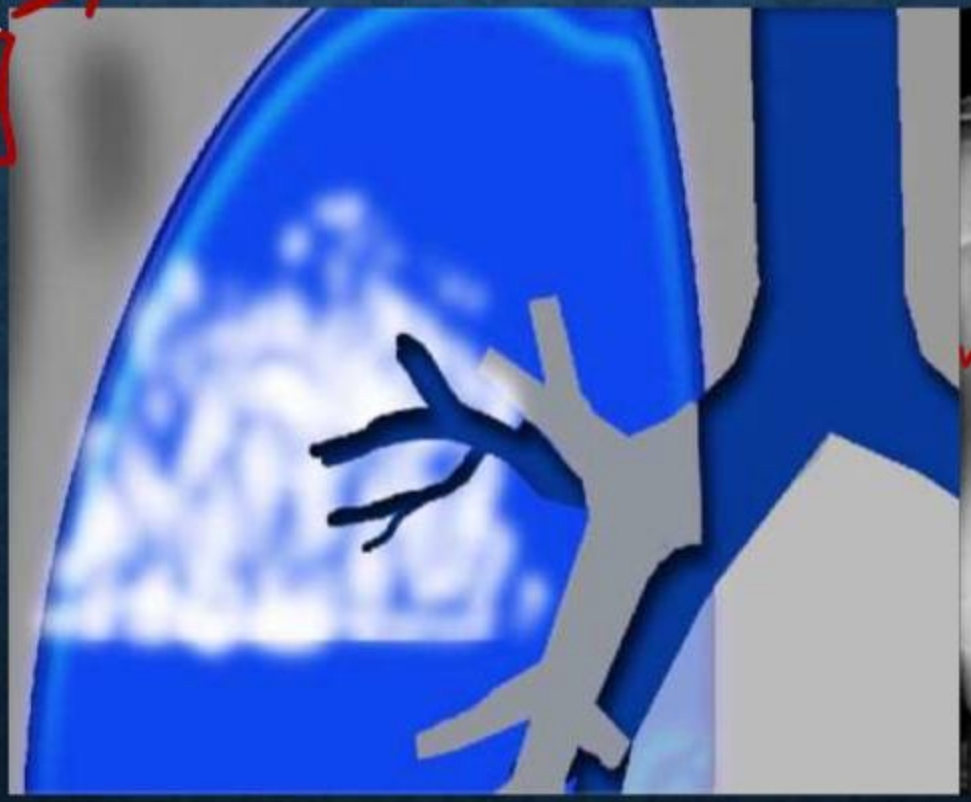
Multiple pulmonary nodules (DDX)

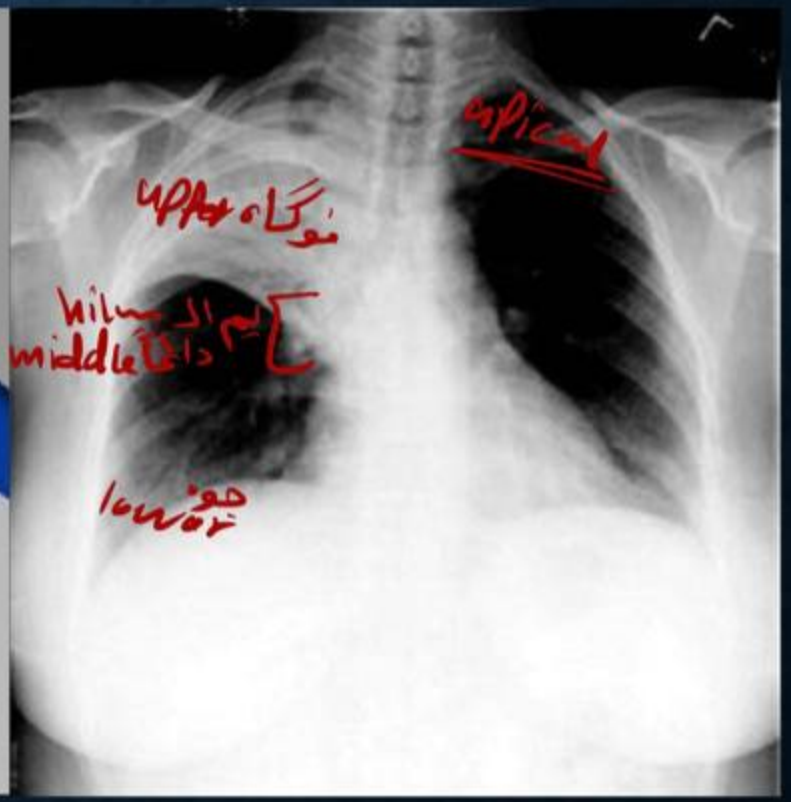
- **Pulmonary metastasis** (most common cause of well defined multiple pulmonary nodules
- Multiple abscesses
- Granulomatous disease (TB, sarcoidosis)
- Fungal infection
- Collagen vascular disease



Consolidation

- **ill-defined or well defined** pulmonary shadow
- **air-bronchogram**
- **NO** volume loss
- Silhouette sign



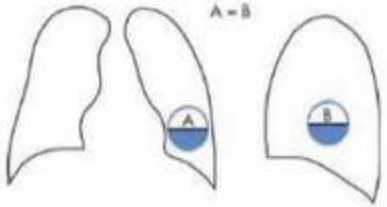
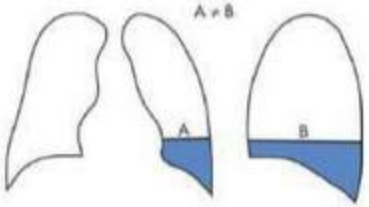


Cavitation

Differential diagnosis of cavitary lung lesion (important)

- Lung abscess.
- Post-primary TB.
- Cavitating neoplasm
- Wegener's granulomatosis.
- Infected emboli

Lung abscess VS empyema

	Abscess	Empyema
Cause	Necrotizing pneumonia (anaerobes, fungus)	Abscess extends to pleura; trauma, surgery
Shape	Round	Elliptical along chest wall
Air-fluid level	A = B	A ≠ B
Margins	Sharp or irregular	Sharp
Wall	Thick	Thin
Lung	Normal position	Displaced
Pleura	Not seen	Split
Vessel/bronchi	Within	Displaced
Treatment	Antibiotics, postural drainage, percutaneous drainage in nonresponders	Percutaneous drainage
		

	Primary TB	Post-primary TB
Location	No lobar predilection	Upper lobe posterior segment or apical segment of lower lobe
appearance	Usually focal	Multifocal patchy involvement
cavitation	No	Frequent(indicate active disease)
LAP	Common	Typically absent
Miliary pattern	yes	yes
Tree in bud	Uncommon	Common (indicate active disease)

Tree-in-bud sign or **pattern** describes the CT appearance of multiple areas of centrilobular nodules with a linear branching pattern. Although initially described in patients with endobronchial tuberculosis, it is now recognized in a large number of conditions.

Signs of lobar atelectasis

Direct signs	Indirect signs
➢ Plate-like or triangular opacity from the collapsed lung itself.	➢ Rib crowding on the side with volume loss.
➢ Displacement of interlobar fissures.	➢ Elevation of the diaphragm.
	➢ Mediastinal shift to the side with volume loss.
	➢ Hilar displacement.
	➢ Overinflation of adjacent or contralateral lobes.

Golden S-sign(reverse S-sign)

- is seen on both PA chest radiographs and on CT scans.

- Seen typically with right upper lobe collapse caused by a central mass obstructing the upper lobe bronchus :

- **Causes are :**

1. Primary bronchogenic carcinoma.
2. Metastasis.
3. Enlarged lymph nodes.
4. Primary mediastinal tumor.

Chest X ray findings of hydrostatic pulmonary edema:

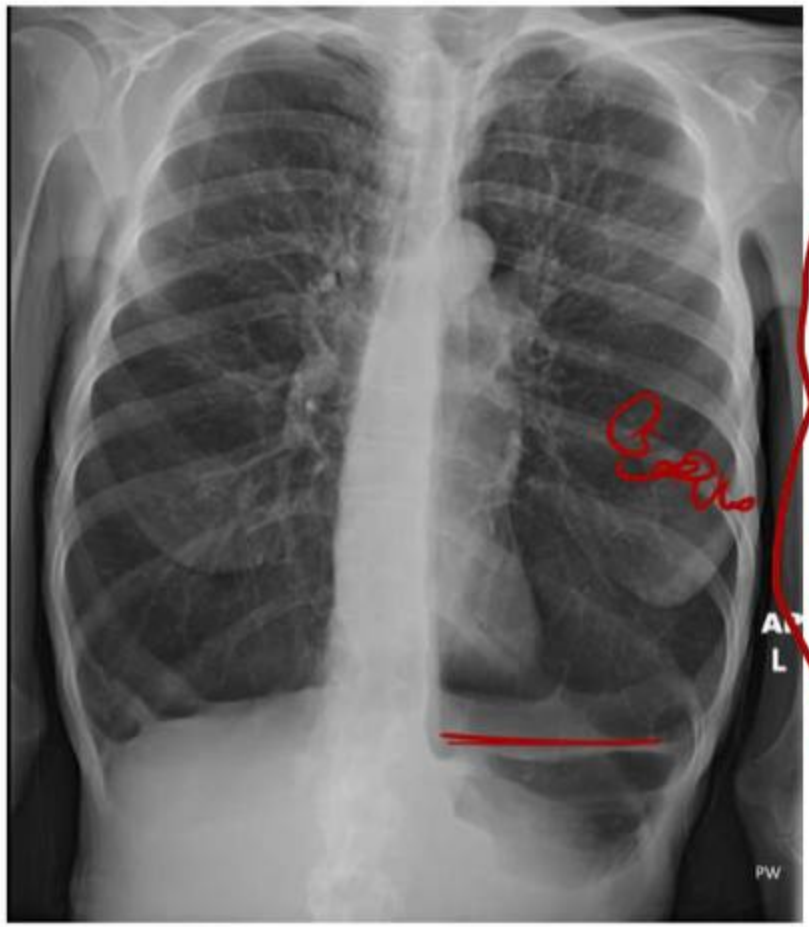
1. Cephalization :redistribution of blood into the upper lobe vessels (upper lobe pulmonary venous diversion)
2. Features of interstitial edema :

✓ Kerley B lines :Thickening of the interlobular septa: they are short parallel lines located at he lung periphery

✓ Peribronchial cuffing : the area around the bronchus appears more prominent on X ray.

✓ Fissural thickening.

3. Air space opacification classically in a batwing distribution(bilateral perihilar opacities)-may have air bronchograms
4. Pleural effusions.
5. Increased cardiothoracic ratio/cardiac silhouette size: useful for assessing for an underlying cardiogenic cause



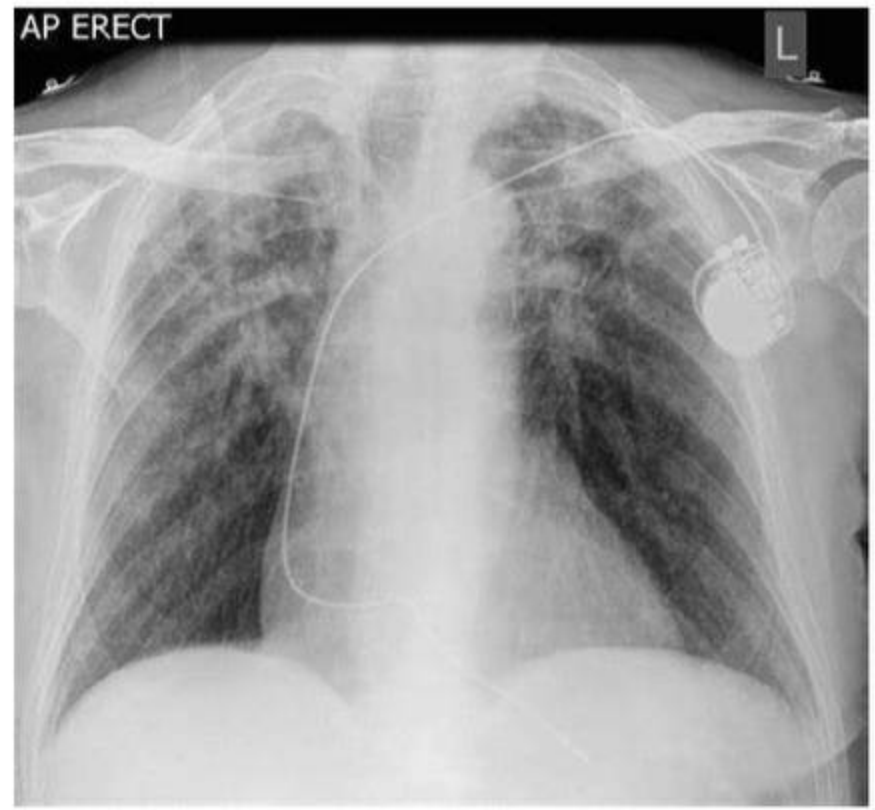
Emphysema X-ray findings include

1. Flattened hemidiaphragms due to overinflation of the lungs
2. Lucent lungs with reduced vascular markings with lung Bullae
3. Narrow or elongated heart shape
4. Widely spaced ribs
5. Saber-sheath trachea narrow & elongated
6. Increased retrosternal airspace(>3 cm)
7. Increased anteroposterior diameter of the chest(>27 cm) barrel chest

Lateral view

Upper lobe fibrotic changes (DDx)

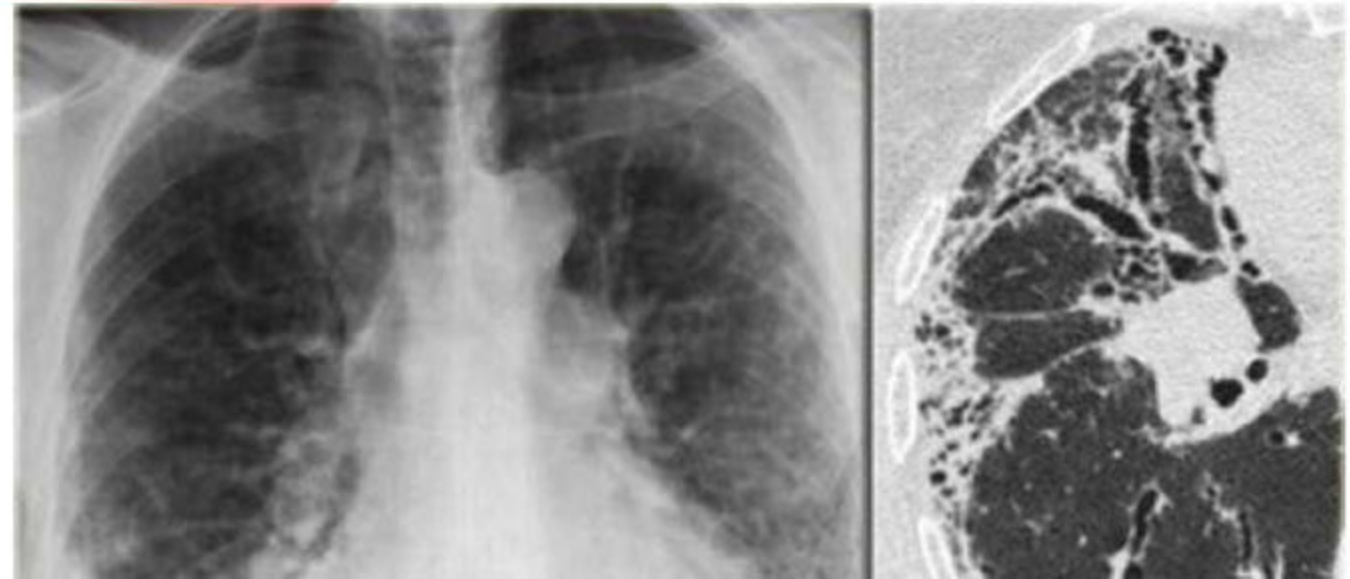
- Sarcoidosis
- TB
- Silicosis and pneumoconiosis.
- Cystic fibrosis.
- Langerhans cell histiocytosis
- Ankylosing spondylitis
- Post-radiation pneumonitis (unilateral)



Pneumoconiosis

Lower lobe pulmonary fibrosis (DDx)

- Usual interstitial pneumonia (idiopathic pulmonary fibrosis) IPF → diagnosis of exclusion
- Collagen vascular disease (RA, scleroderma,…)
- Asbestosis
- Drugs (such as methotrexate)



Radiographic Features

Pleural effusion

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- Homogenous opacity causing obliteration of costophrenic angle.
- Usually the fluid surrounds the lung and is higher laterally than medially (meniscus sign)
- Large effusion causing contralateral mediastinal shift.
- Loculated effusion simulating pleural mass on CXR

Radiological features

Empyema

CXR:

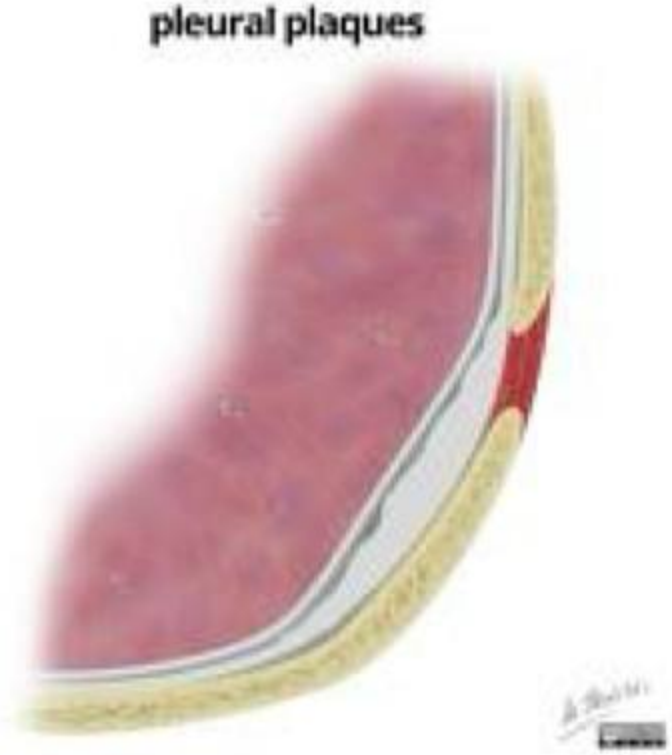
- Lenticular shape collection simulating pleural mass.
- Forming obtuse angle with the chest wall.

CT scan :

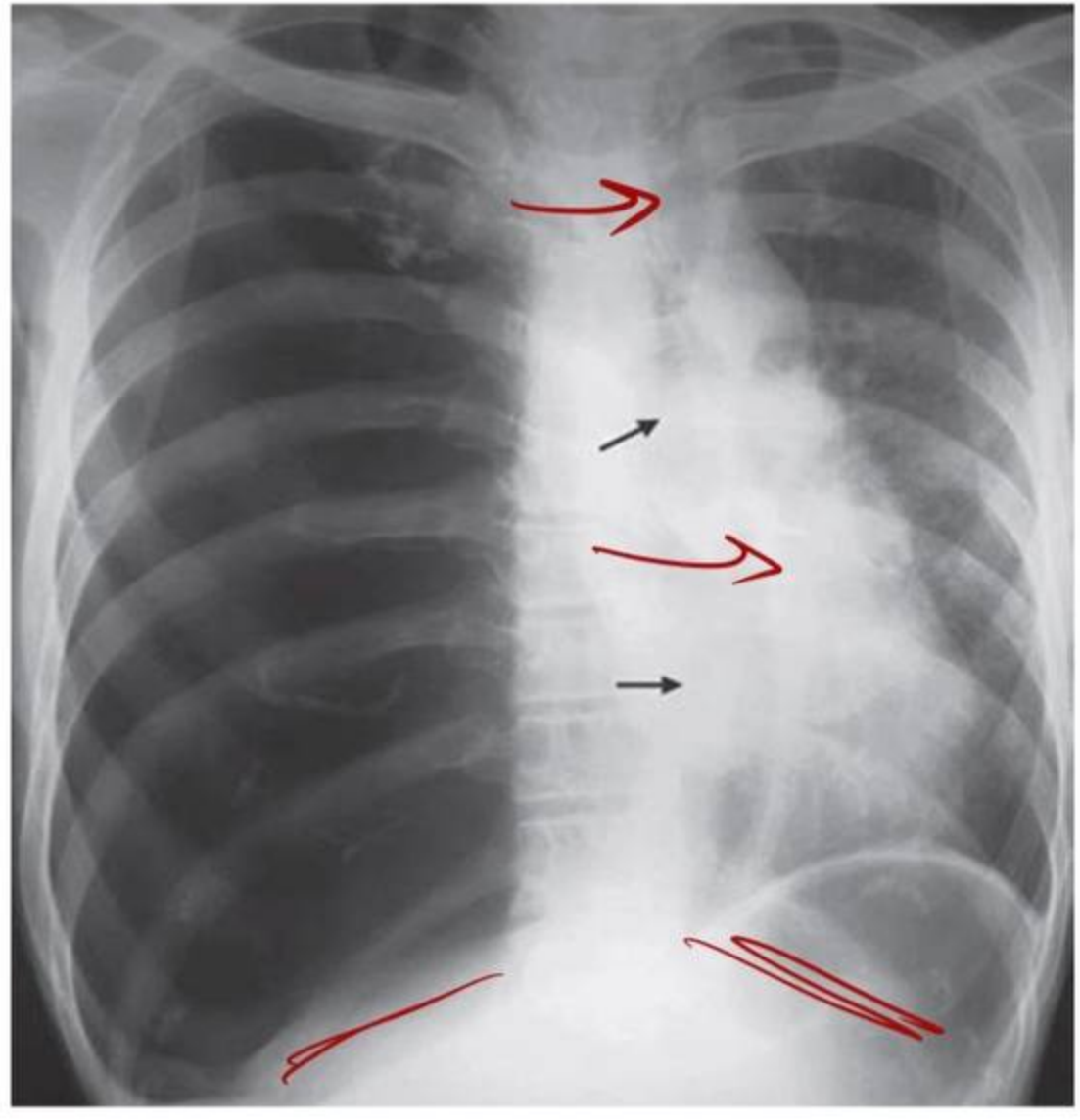
- Lenticular shape fluid collection.
- Thick enhanced wall.
- Split pleura sign (the most sensitive and specific sign differentiating empyema from peripheral lung abscess).

Benign Pleural Thickening

1. Following recurrent inflammation
2. Following pleural empyema.
3. Delayed complication of hemothorax.
4. Asbestos related pleural disease.



Tension Pneumothorax



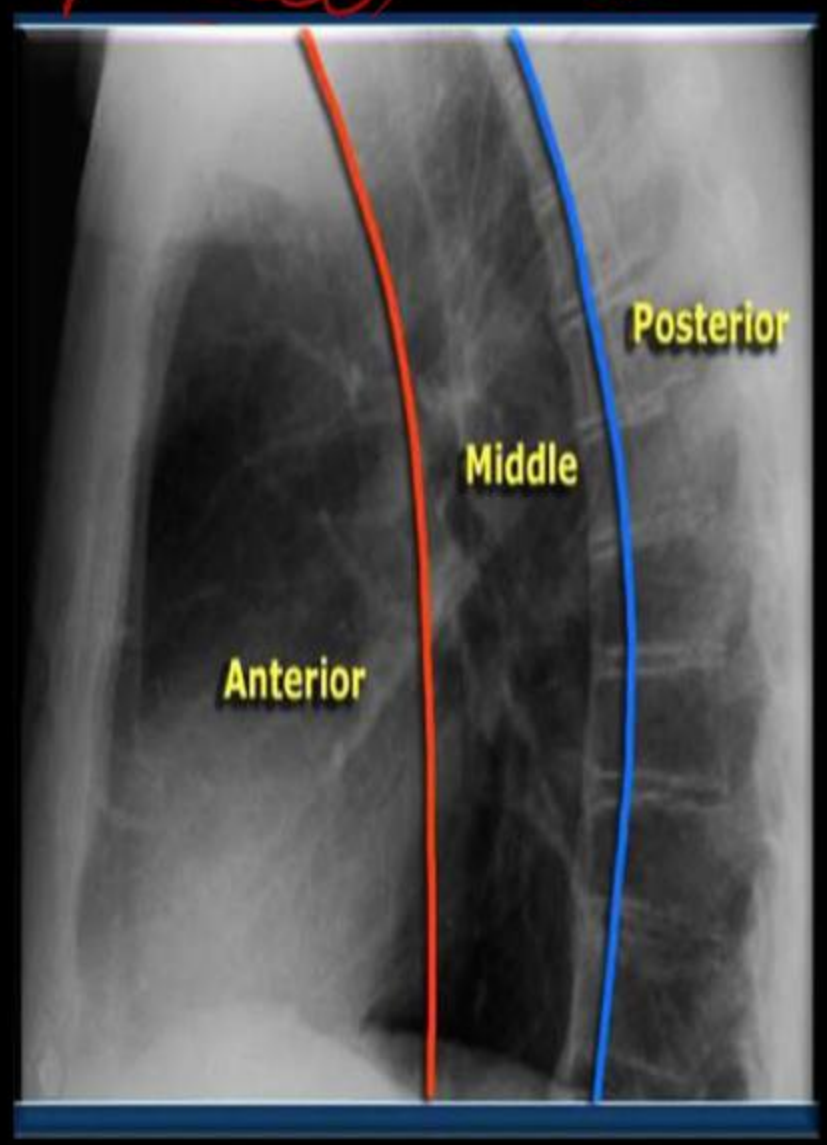
Tension pneumothorax
Contralateral mediastinal shift
Flattening of the hemidiaphragm

IMAGING OF THE MEDIASTINUM

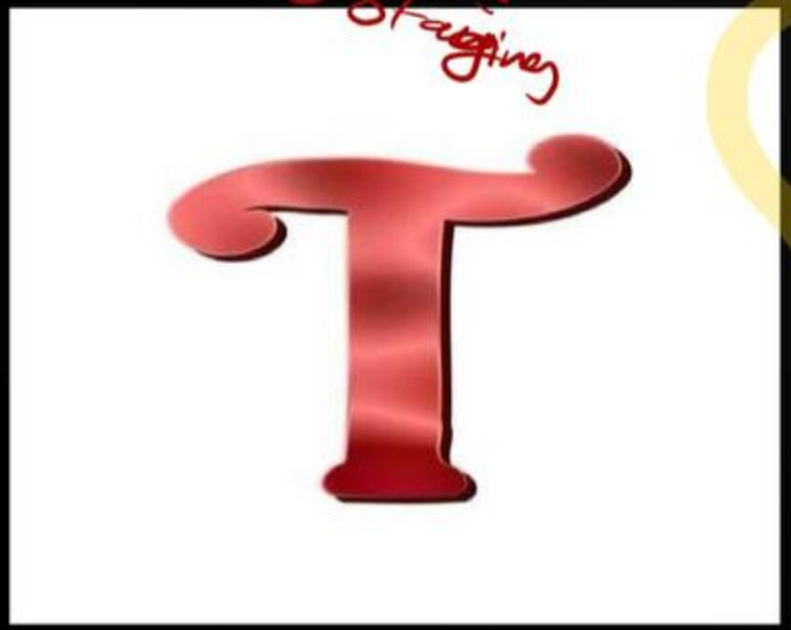
Advantages of CT scan

- **Definitive characterization of mediastinal mass : location, contents (fat/calcium/soft tissue), enhancement and involvement/extension to adjacent structures (operative planning).**
- **Staging of malignant lesions.**
- **Definitive dx of vascular emergencies which require urgent intervention (ruptured aortic aneurysm).**

CT scan protocol : (native + post IV contrast)



ANTERIOR MEDIASTINAL MASS



The DDX of anterior mediastinal mass includes **5** lesions all of which begin with the letter **T.**

- **T**hyroid retrosternal Goiter
- **T**horacic aortic aneurysm
- **T**errible lymphoma
- **T**hymus tumors
- **T**eratoma

MIDDLE MEDIASTINAL MASS

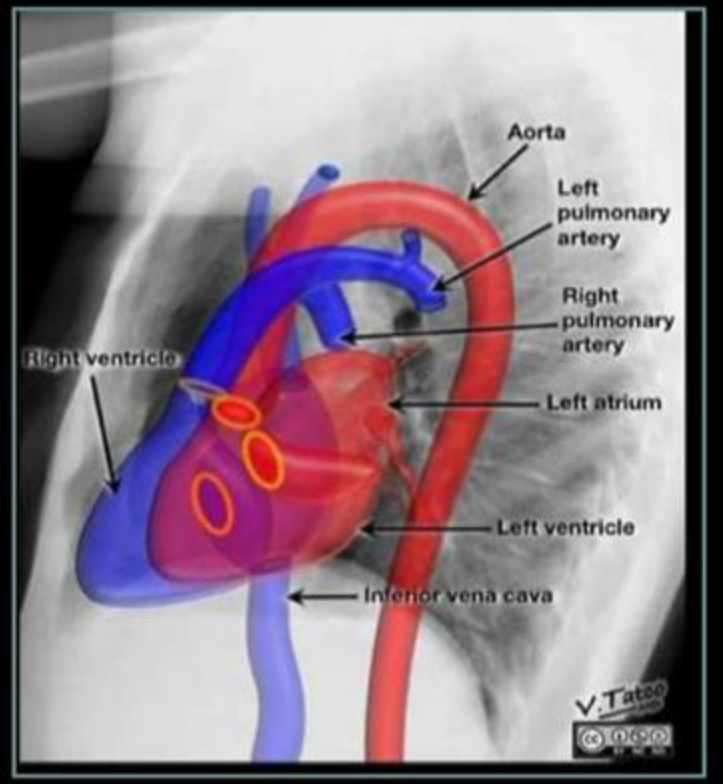
The most important lesions in this compartment are

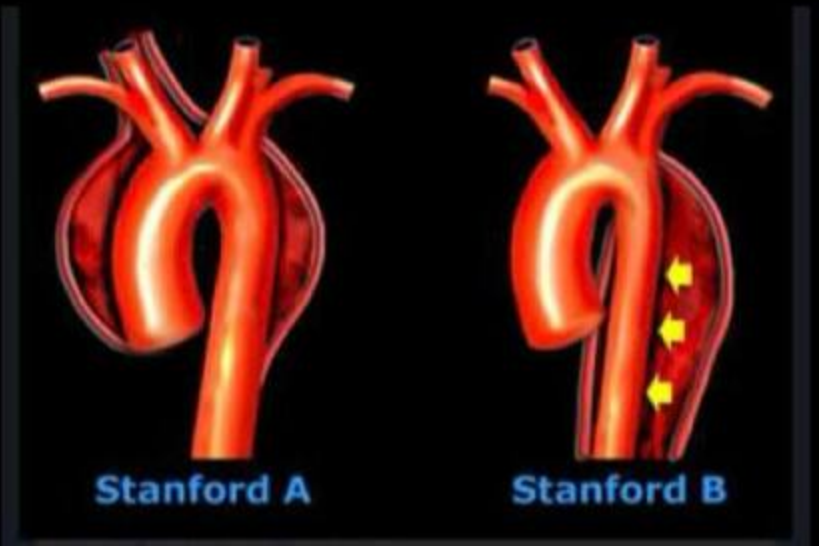
1-Pericardial/bronchogenic cysts 2- Lymphadenopathy.

3- Thoracic aortic aneurysm (which can involve all three mediastinal divisions).

Ruptured thoracic aortic aneurysm is a top emergency which requires early detection and surgical intervention.

CT scan with contrast should be performed to confirm the dx and detect other complications (classification of aneurysm, extension to abdominal aorta/other aortic branches like carotids/coronaries, associated dissection and features of rupture like hemothorax).





POSTERIOR MEDIASTINAL MASS

I. 1- Lymphadenopathy.

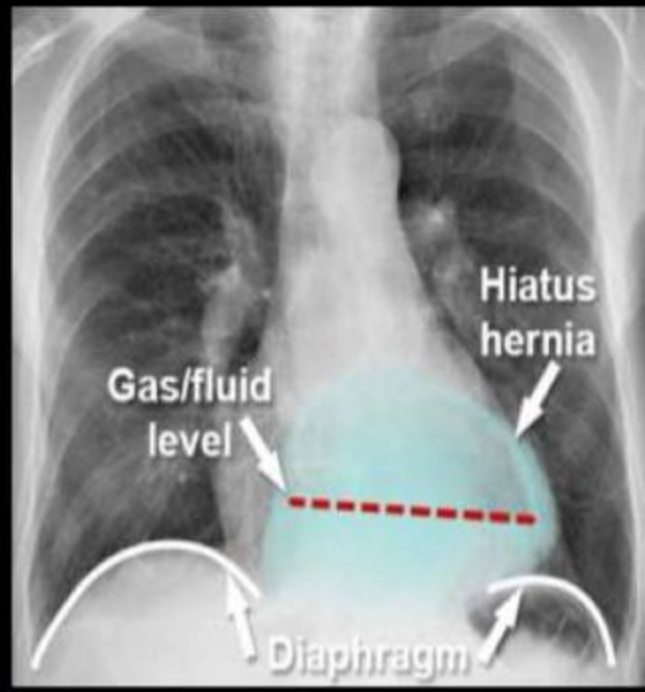
II. 2-Descending thoracic aneurysm.

III. 3- Esophageal/Gastric pathologies : Achalasia and hiatal hernia.

IV. 4- Spine : bone tumors, paraspinal abscess.

V. 5. Neurogenic tumors : the most common posterior mediastinal mass.





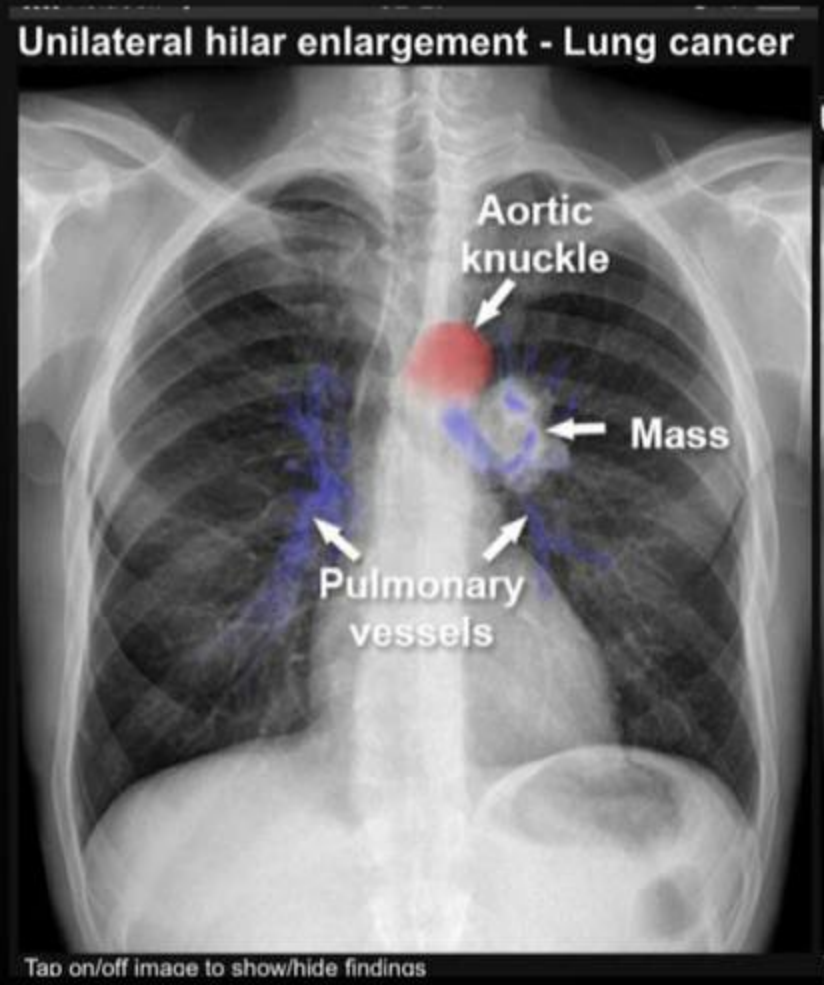
UNILATERAL HILAR MASS

1. Neoplasms including:

- Lung cancer: most common cause.
- Metastasis
- Lymphoma

2. Inflammation : especially Tuberculosis.

3. Large pulmonary arteries: pulmonary hypertension, pulmonary artery aneurysm





BILATERAL HILAR MASSES

A. Neoplasm

- Lymphoma/Leukemia
- Metastases

B. Inflammation

- Sarcoidosis : most common cause of bilateral hilar enlargement.
- Occupational lung diseases

C. Large pulmonary arteries: pulmonary hypertension, pulmonary artery aneurysm



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Carcinoid tumor

Radiological features (Central lesions):

- A single hilar or perihilar mass which is usually well-defined, round or ovoid.
- Typically, the size (2-5 cm).
- Marked homogeneous contrast enhancement due to high vascularity.
- Calcification (usually eccentric) can occur but is not a common feature.

Bronchogenic carcinoma

Risk factors

- Smoking (strongest relationship between smoking and cancer has been established for SCC, followed by adenocarcinoma).
- Radiation, uranium mining.
- Asbestos exposure.
- Genetic predisposition (HLA-Bw44 associated?).

Radiological signs

Primary signs

- Mass > 6 cm
- Speculated / irregular margin
- Cavitation (> 15 mm wall thickness is suggestive of bronchogenic CA).
- Unilateral enlargement of hilum (mediastinal widening, hilar prominence).

Secondary signs

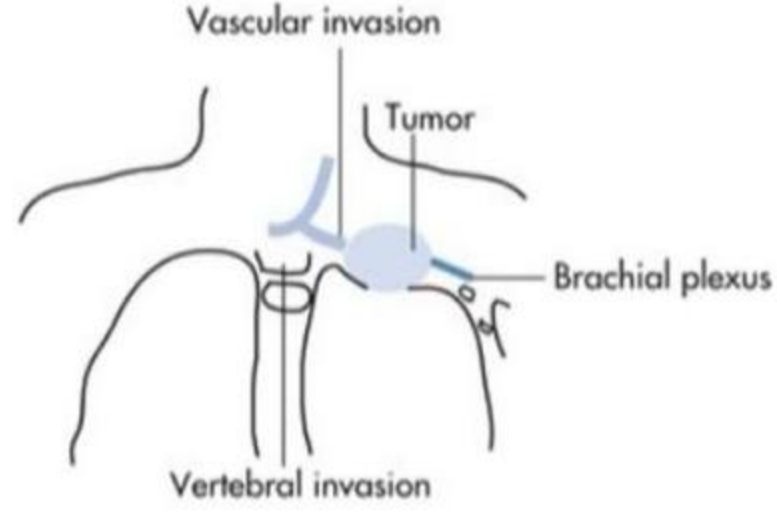
- Atelectasis (Golden inverted S sign in RUL, LUL collapse).
- Obstructive pneumonia.
- Pleural effusion.
- Interstitial patterns: lymphangitic tumor spread.
- Hilar and mediastinal adenopathy.
- Metastases to ipsilateral or contralateral lung.

Radiographic Features

Pancoast tumor
(superior sulcus tumor)

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- Apical mass
- Chest wall invasion
- Involvement of subclavian vessels
- Brachial plexus involvement
- Bone involvement: rib, vertebral body



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Calcified metastases are observed in:

- Bone tumor metastases: Osteosarcoma and Chondrosarcoma
- Mucinous tumors: usually from Ovarian, Colon and Stomach primaries.